

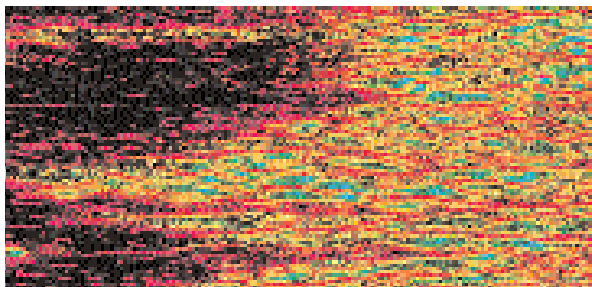
Quantum Computing

Noting the rate at which the number of transistors on a chip is increasing, by 2030 we will be in a position wherein circuits on the microprocessor will be measured on an atomic scale. The next step will then be to program atoms and molecules to do large, high-speed computations—this is quantum computing.

Still at the level of research, quantum computers are believed to boost computation powers to speeds billions of times faster than any silicon-based computer. When introduced into daily life, quantum computers will fuel the growth of technological advancements to greater heights. They work on qubits (quantum bits) which could be in either 0s or 1s. Qubits represent atoms and molecules working in unison as processor and memory. Since a quantum computer can have multiple states

unlike a normal silicon-based computer, it's believed that these computers would be a billion times faster than conventional computers. A 30-qubit quantum computer can achieve speeds equivalent to 10 teraflops (trillions of operations per second). Compare that to supercomputers that are still in the teraflop range.

IBM's Almaden research centre claims to have created the world's fastest quantum computer operating at a speed of 5 qubit. The order-finding problem used in cryptography involves many functions and the quantum computer developed by Almaden solved the functions in a single step—any other computer will require repeated cycles to solve it. Quantum computing being still at a stage of research, it would take many years to come into production.



Simulated airflow over an airplane wing shows the transition from smooth, or laminar, flow (dark areas at left) to turbulence (rippled areas at right)

ing to solve organisational IT issues. IBM is developing the systems software for Globus and the project is called eLiza. This project will serve as the backbone of grid computers and the aim is to provide as many self-manage-

ment technologies as possible which computers need to know when they are on a grid.

More power

The need for faster computation is strongly felt in critical areas of research and development. Researchers need machines that can crunch numbers, formulae and equations that require billions of steps of analysis, predictions, decision-making, formulation, and so on.

Where exactly are supercomputers employed? Formulating astrophysical theories requires analysis of formulae and equations which, in a normal course, would involve decades of human

hours. This is where supercomputing power can be utilised. Britain's biggest supercomputer at the Institute of Computational Cosmology is being used to verify the big bang theory. It analyses trends and existing data and draws a picture of the evolution of the universe. The heavy mathematical and astrophysical calculations are performed by a cluster of 128 Ultra SparcIII processors with a storage capacity of 112 gigabytes and 7 terabytes of RAM.

Areas such as medicine and other biological information systems also need the power of a supercomputer. DNA analysis, for example, would require an unimaginably large number of computations, but supercomputers can ease the task. IBM is working on a five-year project to develop a massive parallel-processing computer that can aid in the study of bio-molecular phenomena such as protein folding. The project called Blue-Gene aims to study bio-molecular phenomena and to develop and understand novel ideas. A petaflop power computer (1015 floating-point operations per second), with something like 50 times the computing power of all supercomputers in the world today, can only be realised through massive parallelism and this is what Blue-Gene tries to achieve. The project has actually evolved from Deep Blue, the computer which won a chess match against Gary Kasparov, the world chess champion in 1997. At the heart of Deep Blue's ability to play chess is its evaluation function. This is an algorithm that measures the 'goodness' of a given chess position. Deep Blue uses 'live' software that can actually generate up to 200,000,000 positions per second while searching for the optimum move. The ASCI White is a cluster of 512 RS/6000 IBM machines, which has a computing power that's 1,000 times more powerful than the Deep Thought. This supercomputer has an architecture that boasts of 160 trillion bytes of storage and a computing power of 12.68 teraflops. The machine can handle Web transactions of every single person in this world in just a minute!

Simulating fluid-flow on a computer involves the computer solving a series of equations, which are used to compute the pressure and velocity of fluid flowing around a body at any point in space, near the body. Application of supercomputers to these equations gave rise to the field of computational fluid dynamics. These equations require

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